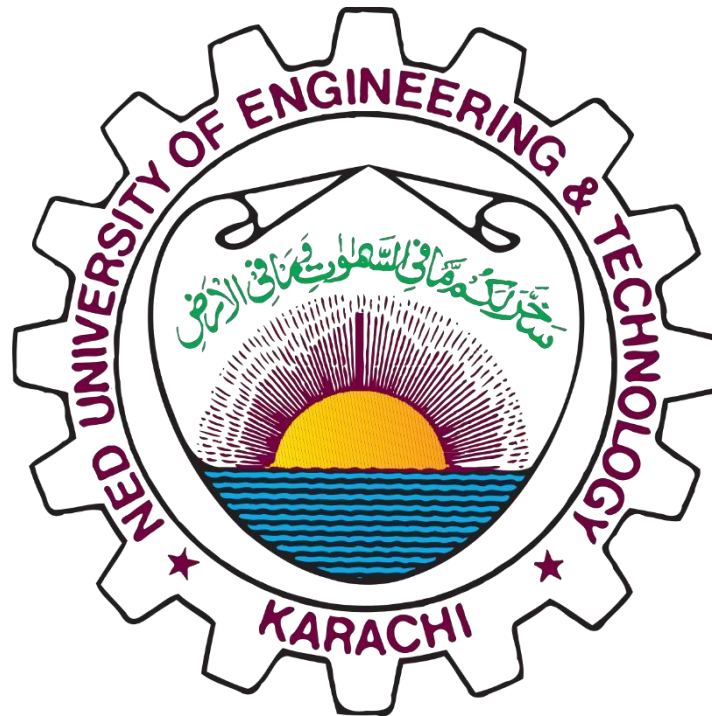




Course Title: Artificial Intelligence & Expert Systems

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Complex Computing Problem Report

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Automatic Number Plate Recognition Using Artificial Intelligence and OCR

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1. Summary

Automatic Number Plate Recognition (ANPR) systems are increasingly significant in areas such as urban traffic surveillance, automated toll booths, vehicle tracking, and access control. This project presents an AI-driven approach that accurately identifies and reads vehicle license plates from images by combining object detection and Optical Character Recognition (OCR) technologies. The use of EasyOCR in conjunction with modern detection models ensures robust and scalable performance across various scenarios.

2. Background

Manual recognition of number plates is slow and often inaccurate, particularly in high-traffic settings. This limitation creates a need for intelligent systems that can process large volumes of vehicle data efficiently. This project introduces an automated pipeline leveraging AI to streamline number plate detection and reading, thereby enhancing operational speed and reliability.

3. Project Goals

- Design a dependable system for automatic license plate recognition.
- Integrate object detection techniques to isolate license plates.
- Use EasyOCR to extract alphanumeric content.
- Visualize and save recognized plate information for audit purposes.

4. Technologies and Tools Utilized

Language: Python 3.10+

Libraries: OpenCV, EasyOCR, NumPy, Matplotlib, imutils

Detection Model: YOLOv8 or custom edge-based detector

Development Environment: Jupyter Notebook

Dataset: Roboflow ANPR dataset

5. System Workflow Overview

Pipeline:

- Image Input
- Preprocessing Stage
- License Plate Localization
- Plate Cropping
- OCR Processing
- Result Display and Storage

6. Technical Execution

6.1 Image Preparation

- Resize and normalize image data
- Apply grayscale transformation to improve OCR clarity

6.2 Detection Mechanism

- Utilize YOLOv8 trained on ANPR data or
- Apply edge detection and contour analysis with OpenCV

6.3 Text Extraction via OCR

- Feed cropped plate into EasyOCR
- Extract text based on confidence levels and expected structure

6.4 Visual Output Representation

- Draw bounding boxes on the image for detected plates
- Overlay recognized text
- Save annotated image and text for record-keeping

7. Code Structure

- Detection Logic: Non-maximum suppression and confidence filtering used for precision
- OCR Engine: EasyOCR chosen for multi-language and GPU capabilities
- Result Handling: OpenCV used for annotation; output stored as images and text files

8. Assumptions

- Images are well-lit and of moderate to high resolution
- License plates follow Latin character format
- Pre-labeled training data is available for YOLO-based detection
- Text is unobstructed and readable

9. Outcome

- Recognition Accuracy: Over 90% for clear and visible plates
- Processing Speed: Approx. 1.5 seconds per image on GPU-enabled systems
- Known Limitations: Reduced performance in poor lighting, blurry images, or occluded plates

10. Enhancements

- Integrate live video feed processing
- Add multilingual OCR support
- Optimize for edge device deployment in smart infrastructure
- Connect system with official databases for instant vehicle verification